



Prüfungsdeckblatt *Examination Cover Sheet*

Studienjahr <i>Academic Year</i>	2023/2024		
Studiengruppe(n) <i>Study Group(s)</i>	BA, IM, LM		
Prüfung <i>Exam</i>	Klausur / Written examination		
Modul(e) <i>Module(s)</i>	Principles of Finance		
Zugelassene Hilfsmittel <i>Permitted aids</i>	Einsprachiges Wörterbuch, nicht-programmierbarer Taschenrechner / monolingual dictionary, non-programmable calculator		
Seiten Aufgabenstellung / Anlagen <i>Pages tasks / appendices</i>	16 Seiten inkl. Deckblatt / 16 pages incl. front page	Formulär (3 pages)	
Prüfungsdatum / -zeit / -dauer <i>Examination date/time/duration</i>	24.06.2024	8 – 9.30 a.m.	90 Minuten Minutes
Prüfer <i>Examiner</i>	Hr. Prof. Prigge (modulverantwortlich)		

Bitte leserlich ausfüllen! Please complete legibly!

Matrikelnummer <i>Matriculation number</i>			
Studiengruppe <i>Study Group</i>			
Anzahl zusätzlicher Seiten Papier <i>No. of additional sheets of paper</i>			
Klausurbeginn /-ende <i>Examination start / end</i>			

Punkteverteilung *Scores*

Aufgabe <i>Task</i>	1	2	3	4	Σ
Maximal <i>Maximum</i>	35	20	20	25	100
Erreicht <i>Achieved</i>					

Note Grade:	Prüfer Examiner:	Unterschrift Prüfer Signature of the examiner:

Punkte-Noten-Schema *Grading Scheme*

Punkte Score	100-95	94-90	89-85	84-80	79-75	74-70	69-65	64-60	59-55	54-50	49-0
Note Grade	1,0	1,3	1,7	2,0	2,3	2,7	3,0	3,3	3,7	4,0	n.b.

Aufgabe Task 1 Investment Decision Rules and WACC (35 Punkte Points)

As a member of the finance department of TipTop AG, you are involved in calculating the company's WACC as well as in preparing a major capital budgeting decision.

a) Your work starts with the WACC. Your colleagues provided the following data:

- cost of equity 10%
- cost of debt 5%
- market value of equity EUR100 million
- market value of debt EUR50 million
- corporate tax rate 33%

a1) Please calculate TipTop's WACC. **(6 points)**

$$\text{WACC} = 50/(100+50) \cdot (1-0.33) \cdot 5\% + 100/(100+50) \cdot 10\% = 7.78\%$$

a2) Please explain the tax advantage of debt in max. **three** sentences. **(4 points)**

The tax advantage of debt is that the interest reduces taxable income. Therefore, the tax payment is reduced.

Dividends do not reduce taxable income and therefore do not reduce tax payments.

Thus, the after-tax cost of debt is below the pre-tax cost of debt.

a3) Corporate tax rates might change in the course of time. Such happened in the U.S., where the corporate tax rate was reduced from 35% to 21% in 2018. What was the effect on the after-tax cost of debt? No calculation required. **(3 points)**

With the corporate tax rate decreases the tax advantage of debt decreased, i.e., the after-tax cost of debt increased.

- b) Turning to capital budgeting, you are part of the team that evaluates the new collection of flip-flop shoes that TipTop might introduce to the market next year. The whole project requires an initial investment outlay of EUR10 million. The following cash surpluses are expected over the next five years (see table; all amounts shown in EUR).

Year	1	2	3	4	5
Cash Flow	5,000,000	4,000,000	2,000,000	1,500,000	750,000

- b1) The CFO asks you to calculate the NPV with the WACC you calculated under a1) as discount rate. Should TipTop AG realize the project? [If you did not calculate a WACC under a1), then use a WACC of 10%.] **(12 points)**

	0	1	2	3	4	5
CF	-10.000.000	5.000.000	4.000.000	2.000.000	1.500.000	750.000
PV	-10.000.000,00	4.639.079,61	3.443.369,54	1.597.406,54	1.111.574,41	515.668,22
NPV		1.307.098,32				

[NPV for discount rate of 10%: EUR844,080.45.]

10 points

NPV is positive → Realize. **2 points**

- b2) In b1), the company WACC of TipTop AG is applied to calculate the NPV and make the investment decision regarding the new flip-flop collection. Under which condition is that permissible? What should be done if the condition is not met? **(4 points)**

Permissible if risk profile of the whole company is similar to that of the new flip-flop collection.

2 points

Project-specific cost of capital. **2 points**

- b3) How would you approach the decision about the flip-flop collection if TipTop disposes for the next fiscal year of several other positive NPV investment projects but cannot, due to a limited budget for investments, realize all of them? Please sketch briefly your proposal and calculate the relevant KPI for the flip-flop project. **(6 points)**

Apply profitability index (PI) to make optimal use of limiting factor, i.e., investment budget.

Rank projects according to their PI and take projects from the top of the ranking until the budget is used up. **4 points**

$PI = 1,307,098.32 / 10 \text{ million} = 0.13$ **2 points**

Company A would like to raise USD12 million to invest in capital expenditures. The company plans to issue three-year bonds with a face value of USD1,000 and a coupon rate of 4% (annual payments). The following table summarizes the yield to maturity for three-year (annual-pay) coupon corporate bonds of various ratings:

Rating	AAA	AA	A	BBB	BB
YTM	3.7%	3.9%	4.0%	4.7%	8.0%

- a) Assuming the bonds will be rated AAA, is it a premium bond or a discount bond? Why? Explain briefly. **(2 points)**

Premium bond because price will be above face value. **1 point**

This is the case because the coupon rate is above market interest rate. **1 point**

- b) Assuming the bonds will be rated AAA, what will the price of the bonds be? **(4 points)**

$$P = 40 \times \frac{1}{0.037} \left(1 - \frac{1}{(1.037)^3} \right) + \frac{1,000}{(1.037)^3} = 111.64 + 896.73 = 1,008.37$$

or

Coupon rate:	4%	Year	1	2	3
Yield to maturity:	3,7%	Cash flow	40	40	1040
Time to maturity:	3	q	1,037	1,037	1,037
Face value:	1000	q^n	1,037	1,0754	1,1152
		PV	38,573	37,197	932,6
		PV total	1008,4		

- c) How much total principal amount of these bonds must Company A issue to raise USD12 million today, assuming the bonds are AAA rated? (Because Company A cannot issue a fraction of a bond, assume that all fractions are rounded to the nearest whole number.) **(4 points)**

Each bond will raise:	1008,3729
Capital to be raised	12000000
number of bonds:	11900,3594
Total principal amount:	11900359,4

- d) What must the rating of the bonds be for them to sell at par? **(3 points)**

For the bonds to sell at par, the coupon rate must equal the yield. Since the coupon is 4.0%, the yield must also be 4.0%, or A rated.

- e) Suppose that when the bonds are issued, the price of each bond is USD896.11.
- e1) What is the IRR of the bond during the three years? Approximation via linear approximation is required. Please calculate two NPVs, one positive and one negative, using the interest rates of 8% and 9%. **(6 points)**

$$r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1)$$

$$NPV = -896.11 + \frac{40}{y} \left(1 - \frac{1}{(1+y)^3} \right) + \frac{1,000}{(1+y)^3}$$

$$NPV_1(8\%) = 0.81; NPV_2(9\%) = -22.67$$

$$r \approx 8\% - \frac{0.81}{-22.67 - 0.81} \cdot (9\% - 8\%) = 8.03\%$$

or

Price of bonds:	896,11	Year	1	2	3
Coupon rate	4%	Cash flow	40	40	1040
Yield to maturity	8,0%	q	1,08	1,08	1,08
Face value	1000	q^n	1,08	1,1664	1,2597
		PV	37,037	34,294	825,59
		PV total	896,92		
		Price of bonds:	896,11		
		NPV ₁ (8%) =	0,8061		

Price of bonds:	896,11	Year	1	2	3
Coupon rate	4%	Cash flow	40	40	1040
Yield to maturity	9,0%	q	1,09	1,09	1,09
Face value	1000	q^n	1,09	1,1881	1,295
		PV	36,697	33,667	803,07
		PV total	873,44		
		Price of bonds:	896,11		
		NPV ₂ (9%) =	-22,675		

r (yield to maturity): 8,03%

- e2) What is the likely rating of the bonds? **(1 point)**

Given a yield of 8.0%, it is likely these bonds are BB rated.

a) Triple-A AG forecasts EUR5.70 as earnings per share (EPS) for year-end. The company's CFO currently assumes earnings to remain constant for all future years. Triple-A's equity cost of capital is 12%.

a1) As there are currently no promising new investment projects, the company has a payout rate of 100%. What is the current stock price? **(4 points)**

Payout rate 100% → $EPS = DIV$

$EUR5.70 / 0.12 = EUR47.50$

a2) What is Triple-A's stock price if the CFO decides to retain 30% of earnings to reinvest in new investment projects with an expected return of 7%? Is the reinvestment decision reasonable? Explain briefly. **(8 points)**

$g = 0.3 \times 0.07 = 0.021$ **(2 points)**

$DIV = EUR5.70 \times 0.7 = EUR3.99$ **(1 point)**

$P_0 = EUR3.99 / (0.12 - 0.021) \approx EUR40.30$ **(3 points)**

Not reasonable, stock price decreases as return on new investment is lower than cost of capital. **(2 points)**

b)

b1) An insider from XYZ AG has just leaked an expected dividend increase of 10% for the running business year in a financial magazine. Officially, the increase will be announced a few weeks later during the annual meeting of the company. The stock price of XYZ AG immediately increases by 10% on the publication date of the leak in the financial magazine. Please comment briefly on the prevailing form of market efficiency here. **(5 points)**

We can assume semi-strong form of market efficiency here. Semi-strong form implies that all publicly available information is incorporated into stock prices and prices react only to new public information. It seems that the dividend increase is higher than extant market expectation. As the expected dividend increase becomes public, the stock will immediately increase accordingly.

b2) A few years later XYZ AG has a new ongoing business venture. In a few days the management team will hold a conference with investors to present a first outlook on the venture's prospects. If the venture is a failure, the severe amount of sunk costs will let the stock price crumble to EUR20. If the venture should be a promising success, it will lift XYZ AG to the top-class of their industry resulting in a share price of EUR85. The stock currently trades at EUR65. What is the expected probability of success for the venture deemed by financial markets (calculation required)? **(3 points)**

As the current stock price is closer to the success case than to the failure case, the market price implies a probability of success higher than 50%. **2 points**

$$85 \times X + 20 \times (1 - X) = 65$$

$$85X + 20 - 20X = 65$$

$$65X = 45$$

$X \approx 0.69 \rightarrow$ Financial markets expect the venture to be successful with a probability of 69 %.

1 point

- a) Please explain the difference between systematic and unsystematic risk. **(4 points)**

Systematic risk is market risk, thus, affecting the whole economy.

Unsystematic risk is not general, but specific, associated with a single firm or a specific industry.

The unsystematic risk can be diversified away in portfolios.

- b) Please explain why there is no variable to represent the unsystematic risk in the SML formula of the CAPM. **(4 points)**

Unsystematic risk can be diversified away.

Equity investors cannot expect extra returns for risks they do not have to take.

- c) Your team is analyzing the stock of a mining company with above-average volatility (standard deviation). Your colleague believes that such a stock has also a high beta because “a stock with high unique risks always has also a high systematic risk”. Is he right? Please explain your answer. **(5 points)**

My colleague is wrong. A company could have high unique risks associated with its business model. Mining companies are facing for example the risks of the gold price volatility on the sales side, and political risks or risks with regard to their gold mines on the production side. The production side risks are usually firm-specific or sector specific. Hence, they lead to higher volatility, but not to a higher beta. On the sales side, gold is meant to be a “safe haven” in crisis times. So, the gold price increases when the economy declines and vice versa, also contributing to a low beta. Summarizing, companies might have a big exposure to unique risks, but only a minor exposure to systematic risk.

- d) As of now, you hold only stocks of TipTop AG. As a member of the finance department, you know about the beneficial effects of diversification, so you intend to buy shares from a second company. The stocks of TopTip AG seem to be an attractive addition. Therefore, your new portfolio consists of TipTop AG and TopTip AG, both having a weight of 50%. You have the following forecasts about expected returns for next year, where you assume that each state of the economy is equally likely. The table below shows the relevant figures:

	Recession	Normal	Boom	Expected Return	Standard Deviation
TipTop AG	-9.00%	8.00%	21.00%	6.67%	12.28%
TopTip AG	10.00%	4.00%	-5.00%	3.00%	6.16%

- d1) What is the expected return of the portfolio? **(4 points)**

- d2) What is the standard deviation of the portfolio? **(5 points)**

	Recession	Normal	Boom	Exp. Return	SD
TipTop AG	-9,00%	8,00%	21,00%	6,67%	12,28%
TopTip AG	10%	4%	-5%	3,00%	6,16%
Portfolio	0,50%	6,00%	8,00%	4,83%	3,17%

- d3) Is the diversification effect caused by the addition of TopTip to your portfolio big? Please explain briefly. Is there a quick indicator to estimate the size of the diversification effect? **(3 points)**

Big diversification effect because of low correlation between both stocks (one counter-cyclical, one pro-cyclical). **2 points**

Quick indicator: standard deviation in case of perfect positive correlation = no diversification effect at all = $0.5 \cdot 12.28\% + 0.5 \cdot 6.16\% = 9.22\%$. Actual standard deviation of 3.17% is much lower. **1 point**

Formulary – Appendix to Exam

Chapter 1

Future Value of a Cash Flow: $FV_n = C \times (1 + r) \times (1 + r) \times \dots \times (1 + r) = C \times (1 + r)^n$

Present Value of a Cash Flow: $PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n}$

Net Present Value (NPV): compares the present value of cash inflows (benefits) to the present value of cash outflows (costs).

Present Value of a Perpetuity (first payment at time 1): $PV(C \text{ in perpetuity}) = \frac{C}{r}$

Present Value of an Annuity: $PV(\text{annuity of } C \text{ for } N \text{ periods with interest rate } r) = \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N}\right)$

Future Value of an Annuity:

$$\begin{aligned} FV(\text{annuity}) &= PV \times (1 + r)^N \\ &= \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N}\right) \times (1 + r)^N \\ &= C \times \frac{1}{r} ((1 + r)^N - 1) \end{aligned}$$

Present Value of a Growing Perpetuity: $PV(\text{growing perpetuity}) = \frac{C}{r - g}$

Present Value of a Growing Annuity: $PV = C \times \frac{1}{(r - g)} \left(1 - \left(\frac{1 + g}{(1 + r)}\right)^N\right)$

Approximate Internal Rate of Return (IRR) via Interpolation
(with $NPV_1 > 0$ and $NPV_2 < 0$): $r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1)$

Chapter 2

Profitability Index: $\text{Profitability Index} = \frac{\text{Value Created}}{\text{Resource Consumed}} = \frac{NPV}{\text{Resource Consumed}}$

Chapter 3

Coupon Payment: $CPN = \frac{\text{Coupon Rate} \times \text{Face Value}}{\text{Number of Coupon Payments per Year}}$

Price of a Zero-Coupon Bond (FV=face value): $P = \frac{FV}{(1 + YTM)^n}$

Yield to Maturity of an n-Year Zero-Coupon Bond: $YTM = \left(\frac{FV}{P}\right)^{\frac{1}{n}} - 1$

Yield to Maturity of a Coupon Bond: discount rate y , that equates the PV of the bond's future cash flows with its price (=IRR): $P = CPN \times \frac{1}{YTM} \left(1 - \frac{1}{(1 + YTM)^N}\right) + \frac{FV}{(1 + YTM)^N}$

Chapter 4

Dividend-discount model (1 year): $P_0 = \left(\frac{\text{Div}_1 + P_1}{1 + r_E} \right)$

Total return, capital gain, dividend yield: $r_E = \frac{\text{Div}_1 + P_1}{P_0} - 1 = \underbrace{\frac{\text{Div}_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$

Dividend-discount model (2 years): $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2 + P_2}{(1 + r_E)^2}$

Dividend-discount model (n years): $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \dots + \frac{\text{Div}_N}{(1 + r_E)^N} + \frac{P_N}{(1 + r_E)^N}$

$$P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \frac{\text{Div}_3}{(1 + r_E)^3} + \dots = \sum_{n=1}^{\infty} \frac{\text{Div}_n}{(1 + r_E)^n}$$

DDM, constant growth: $P_0 = \frac{\text{Div}_1}{r_E - g} \quad r_E = \frac{\text{Div}_1}{P_0} + g$

DDM, changing growth rates: $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \dots + \frac{\text{Div}_N}{(1 + r_E)^N} + \frac{1}{(1 + r_E)^N} \left(\frac{\text{Div}_{N+1}}{r_E - g} \right)$

Change in Earnings = New Investment × Return on New Investment

New Investment = Earnings × Retention Rate

Earnings Growth Rate = $\frac{\text{Change in Earnings}}{\text{Earnings}} = \text{Retention Rate} \times \text{Return on New Investment}$

$g = \text{Retention Rate} \times \text{Return on New Investment}$

Dividend: $\text{Div}_t = \frac{\text{Earnings}_t}{\underbrace{\text{Shares Outstanding}_t}_{\text{EPS}_t}} \times \text{Dividend Payout Rate}_t$

Chapter 5

Expected (Mean) Return: Calculated as a weighted average of the possible returns, where the weights correspond to the probabilities.

$$\text{Expected Return} = E[R] = \sum_R P_R \times R$$

Variance: The expected squared deviation from the mean.

$$\text{Var}(R) = E[(R - E[R])^2] = \sum_R P_R \times (R - E[R])^2$$

Variance Estimate Using Realized Returns:

$$\text{Var}(R) = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2$$

Standard Deviation: The square root of the variance.

$$SD(R) = \sqrt{\text{Var}(R)}$$

$$\begin{aligned} \text{Portfolio rate of return} &= (\text{fraction of portfolio in asset 1} \times \text{rate of return on asset 1}) \\ &+ (\text{fraction of portfolio in asset 2} \times \text{rate of return on asset 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in asset } t \times \text{rate of return on asset } t) \end{aligned}$$

$$\begin{aligned} \text{Portfolio beta} &= (\text{fraction of portfolio in stock 1} \times \beta \text{ of stock 1}) \\ &+ (\text{fraction of portfolio in stock 2} \times \beta \text{ of stock 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in stock } t \times \beta \text{ of stock } t) \end{aligned}$$

$$\text{Risk premium on the market} = r_m - r_f$$

$$\text{Risk premium on any asset } i = r_i - r_f$$

$$\text{SML equation: } r = r_f + \beta(r_m - r_f)$$

Chapter 6

$$r_{\text{assets}} = \frac{\text{total income}}{\text{value of investments}} = \frac{(D \times r_{\text{debt}}) + (E \times r_{\text{equity}})}{V}$$

$$r_{\text{assets}} = \frac{D}{V} \times r_{\text{debt}} + \frac{E}{V} \times r_{\text{equity}}$$

$$\text{After-tax cost of debt} = r_{\text{debt}} \times (1 - T_c).$$

$$\text{WACC} = \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} \right] + \left[\frac{E}{V} \times r_{\text{equity}} \right]$$